Correspondence

Efficacy and safety of limbal anaesthesia for clear cornea phacoemulsification

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Editor,

We read with special interest the article 'Efficacy and safety of limbal anaesthesia for clear cornea phacoemulsification' by Cagini et al. (2004). This is surely an innovative technique located between topical anaesthesia and no-anaesthesia cataract surgery. We are interested in clarifying a few aspects of the procedure. Firstly, how much patient co-operation is required when a lidocaine swab is placed on a non-anaesthetized conjunctiva? What technique is used to stabilize the globe at the time of initial entry with the keratomes? Topical anaesthetic was used prior to the instillation of betadine drops into the cul-de-sac. This topical anaesthesia may have been helpful in the initial placement of the eye speculum and the lidocaine swab. This technique surely reduces the need for frequent instillation of topical anaesthetic drops, but the technique cannot be described as purely limbal anaesthesia.

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In response to comments on limbal anaesthesia for clear cornea phacoemulsification

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Editor,

First of all, we would like to thank correspondents Rai and Singh Sethi for their appreciative comments regarding this innovative technique. However, we would like to point out that, in our opinion, the proper placement of limbal anaesthesia (LA) would be as an evolution of topical anaesthesia. In fact, the patient's level of analgesia and the safe surgery achieved with this method are similar to those obtained with topical anaesthesia. This is demonstrated by a randomized, double-blind study we have recently completed and which we plan to publish shortly.

With LA, patient co-operation is excellent because we instil one drop of anaesthetic solution outside the operating room immediately before the patient is brought in for surgery. This permits sufficient surface anaesthesia for application of the lid speculum and for application of the sponge soaked in preservative-free lidocaine can be applied for 45 seconds. Limbal anaesthesia provides of analgesia of the entire circumference of the limbus and of every quadrant of the cornea, similar to that obtained with topical anaesthesia. Moreover, LA ensures adequate iris and ciliary body anaesthesia. The precise mechanism of this anaesthesia is not completely understood: corneal anaesthesia results from blocking the pericorneal and annular plexus, whereas it is likely that the penetration of anaesthetic agents through the corneal and conjunctival–scleral route results in iris and ciliary body anaesthesia (Ahmed & Patton 1985; Shoenwald et al. 1997). In fact, in our experience, the patient does not feel any pain during contact with the main point of entry or during the side-port incisions, and it is possible to execute the entire procedure safely. Therefore, LA completely eliminates the need for repeatedly instilling the anaesthetic solution, and must be preceded by a single drug application in order to instil the disinfectant and apply the blepharostat.

A similar experience was presented by Lanzetta et al. (2000), who proposed perilimbal topical anaesthesia without the addition of anterior chamber irrigation with lidocaine in routine clear corneal cataract surgery; they demonstrated that, with this new anaesthesia, it is possible to achieve a sufficient level of anaesthesia to perform surgery safely.

Consequently, the main feature of limbal anaesthesia is the fact that it further simplifies preoperative procedures, and that the surgeon can be confident that an adequate quantity of anaesthetic has been administered, avoiding the toxic effects caused by multiple or excessive administration of the drug. In our operating rooms, LA has now become the anaesthesia of choice in surgery of the anterior segment. We also use it routinely in trabeculectomies and are conducting a randomized double-blind study comparing LA with peribulbar anaesthesia in phacotrabeculectomy procedures.

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Survival after primary enucleation of largest uveal melanomas

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Editor,

We read with great interest the article by Isager et al. (2004), which studied the survival rates of patients in Århus County, Denmark treated by primary enucleation for uveal melanoma during the period 1955–2000. The authors reported a 5-year melanoma-specific survival probability of 70% (95% confidence interval 64–75%).

The population analysed was heterogeneous because the study referred to melanomas enucleated both before and after the advent of conservative treatments (1988 in Århus County). The management of uveal melanoma has since been changed by conservative radiotherapies (brachytherapy or proton-beam therapy). Nowadays, primary enucleation is performed on melanomas only when there is no hope of regaining useful vision (i.e. the largest tumours and melanomas with initial complications such as neovascular glaucoma) (De Potter 2002). Large tumoral size, which determines the choice of primary enucleation, represents the most significant clinical prognostic factor for melanoma-related death. In this way, melanomas currently treated by primary enucleation seem to differ substantially from melanomas treated by conservative treatments and also from melanomas enucleated before the advent of radiotherapies.

For this reason it is regrettable that the survival rate of patients enucleated during the period 1988–2000 was not specified in the article by Isager et al. (2004). Information on melanoma-specific survival rates for patients treated prior to the advent of eye-preserving therapies is interesting only in a historic sense; it is not useful in daily practice. Knowledge of melanoma-specific survival rates for primarily enucleated patients after the advent of eye-preserving therapies is quite essential to current practice in terms of advising patients on courses of treatment.

We studied melanoma-specific survival rates of patients treated by primary enucleation after the advent of conservative radiotherapies in our clinic in Lyons, France. Since 1991, the complete portfolio of eye-preserving treatments used for uveal melanoma has been available in our department. Forty patients, representing 8% of patients diagnosed and followed up in our department during the period 1991–2002, have undergone primary enucleation. Our 5-year melanoma-specific survival rates for the period are 31.45% (standard error [SE] 8.04%) after primary enucleation (data not yet published), 78.1% (SE 3.7%) after proton-beam irradiation and 82% (SE 2.7%) after ruthenium brachytherapy (Kodjikian et al. 2004; Rouberol et al. 2004). These results particularly demonstrate the pejorative prognosis for the largest tumours, for which primary enucleation remains the reference treatment.

The majority of published reports have analysed melanoma-specific survival only in populations that have been treated by conservative radiotherapies, without mentioning the proportion of melanomas that were primarily enucleated. Not taking into account the enucleated population biases any comparison with studies that pre-date the advent of eye-preserving therapies. Moreover, it leads to an implicit overestimation of the melanoma-specific survival rate of the entire population.

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Survival after enucleation and radiation brachy therapy for choroidal and ciliary body melanomas

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Editor,

We would like to thank correspondents Gambrelle et al. for their interest in our paper 'Prognostic factors for survival after enucleation for choroidal and ciliary body melanomas' (Isager et al. 2004).

The proportion of patients treated by enucleation versus those treated by eye-preserving therapies varies between melanoma centres, a fact that should be accounted for in the interpretation of published survival data.

During the period 1988–2000, 61 patients with large choroidal/ciliary body melanomas were enucleated, with a 5- and 10-year melanoma-specific survival of 67% (95% CI 52–78%) and 42% (95% CI 22–61%), respectively. During the same period, 60 patients with small and medium-sized tumours were treated by radiation brachytherapy (mainly Ru/Rh-106) with a 5- and 10-year melanoma-specific survival of 85% (72–92%) and 73% (54–85%), respectively. One further patient was treated by local tumour resection. As expected, the survival probability depended on the tumour size.

We hope that these survival data will be of use to ophthalmologists who need to advise patients in the future.

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